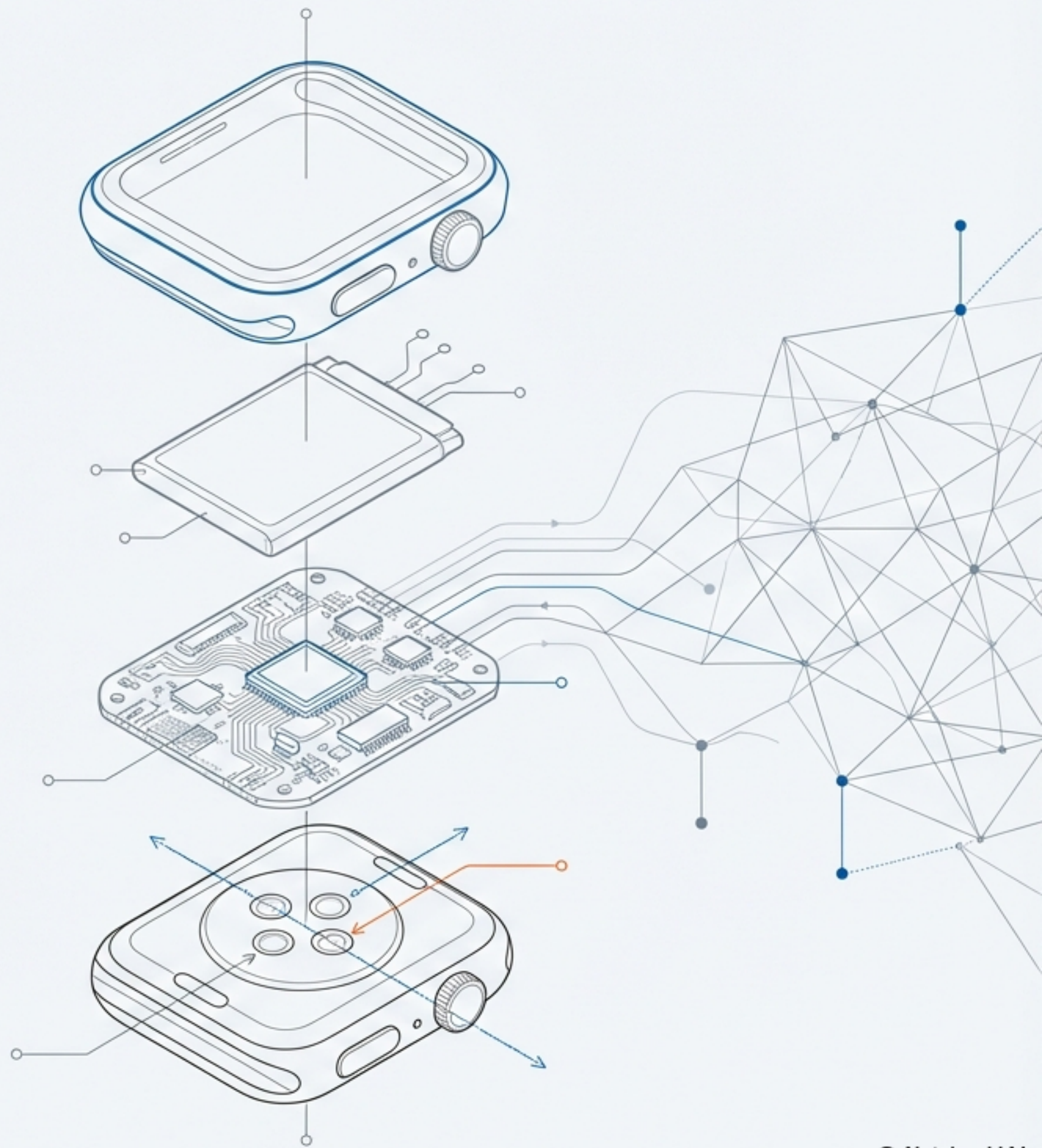


The Fairness Audit Playbook for Edge-Based Healthcare AI

Integrating Fairness, Privacy, and Contextual Robustness in Wearable Systems under GDPR.



Moving Beyond 'Accuracy' in Wearable Health

Executive Summary

The Context

Edge intelligence drives modern healthcare wearables.

However, standard benchmarks hide systemic risks where privacy mechanisms (Federated Learning & Differential Privacy) actively degrade fairness for marginalized groups.

The Problem

- **Historical Bias:** Sensors validated on young/healthy cohorts fail on elderly/impaired users.
- **Hardware Heterogeneity:** Low-cost devices with aggressive battery saving produce lower quality data.
- **The Hidden Trade-off:** Privacy tools disproportionately harm minority performance.



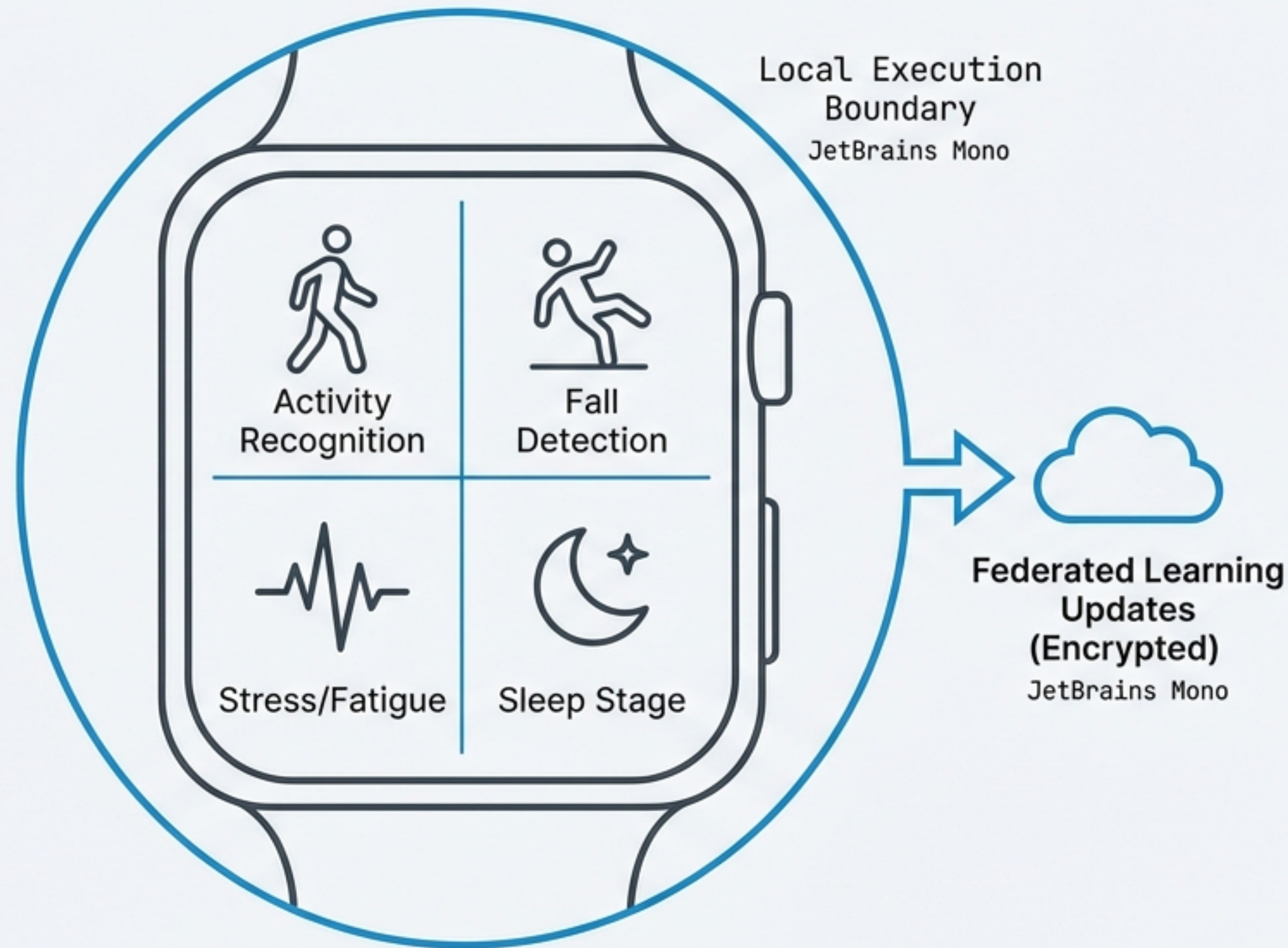
The Solution

A unified Playbook treating Fairness, Privacy, and Robustness as joint system properties.

The output is a repeatable workflow aligning engineering metrics with GDPR obligations (Art. 5 & 25).



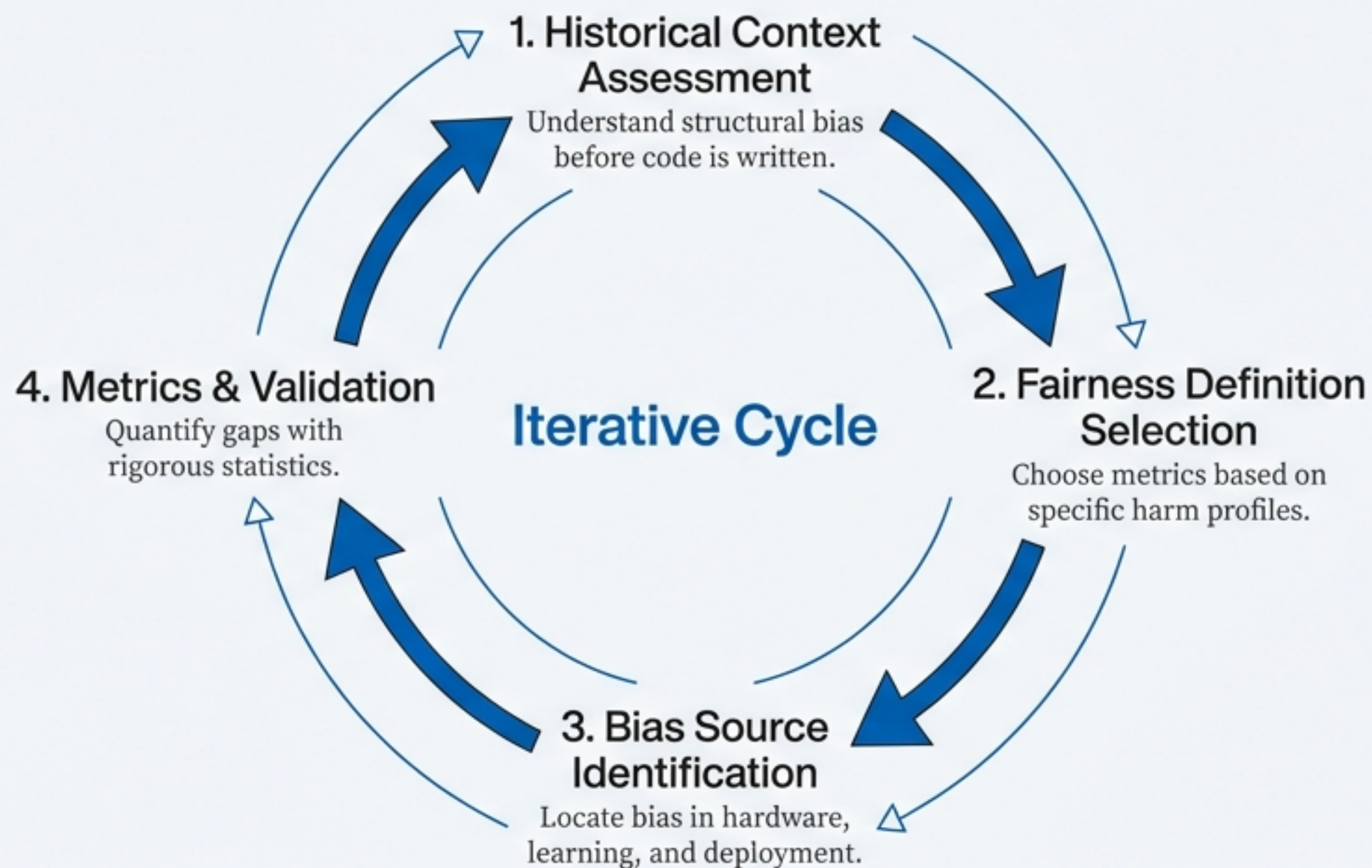
The Audit Target: Edge-Based Health Monitoring



Technical Mechanics

- Local Inference Execution
- Federated Learning Updates
- Differential Privacy (Client/Aggregation)

A Unified Workflow for Responsible AI



This framework replaces fragmented assessments with a single pipeline.
Fairness requires understanding the physical context of the user.

Fairness Issues Are Structural, Not Just Algorithmic

Model Accuracy Metrics →

Design Bias: Sensors validated on young/healthy cohorts fail on elderly/impaired users (gait, skin tone).

Physiological Baselines:
Variations in heart rate/sleep by age/gender treated as noise.

Data Stratification:
Low-SES users = Cheaper devices
= Lower quality data.

AUDIT TARGETS:

- Elderly Users
- People with Mobility Impairments
- Women
- Low-SES Populations

LEGAL DISCLAIMER: These are decision-support/wellness tools only. System does not perform diagnosis, treatment, or automated emergency response.

One Definition Does Not Fit All Tasks

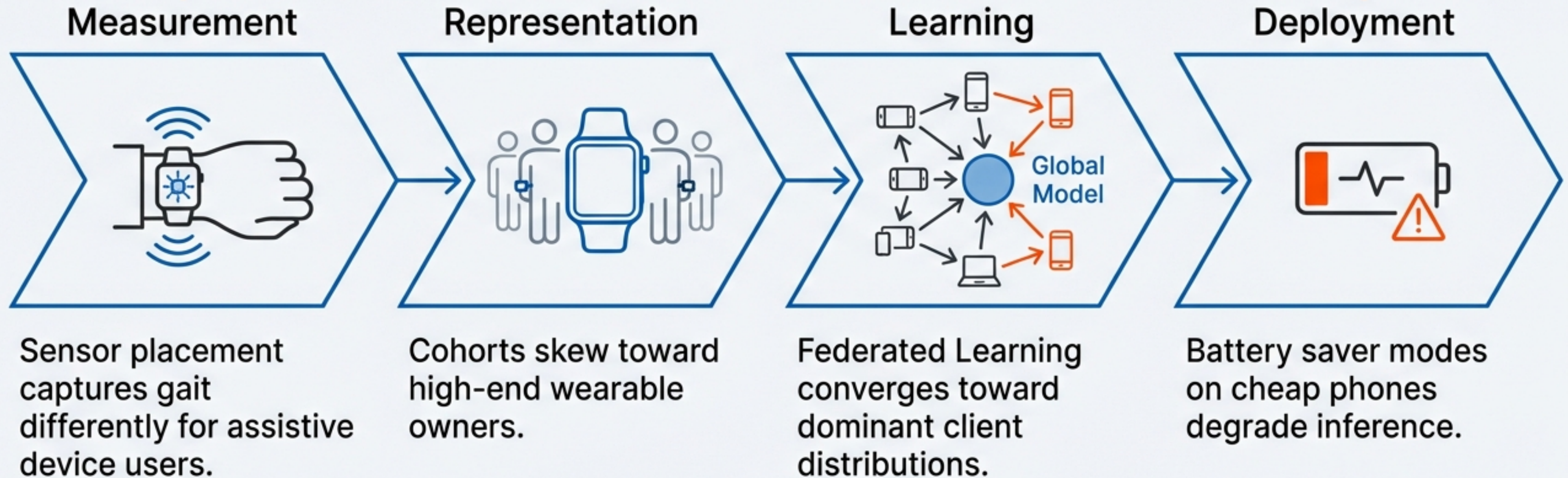
Mapping Fairness Definitions to Module Risk

Module	Fairness Definition	Rationale
Activity Recognition	Equal Opportunity	Do not miss the activity. False negatives hurt users with non-standard gait.
Fall Detection	Equalised Odds	Balance required. False negatives are dangerous; False positives cause panic.
Stress/Fatigue	Individual Fairness	Similar physiological patterns must yield similar outputs.
Sleep Staging	Stage-Specific Recall	Must not miss clinically relevant stages (e.g., deep sleep).

 **Note: ‘Demographic Parity’ is explicitly rejected to avoid masking legitimate physiological differences.**

Where Does Bias Enter the Edge Pipeline?

Bias is a physical and hardware phenomenon, not just software.



The Myth of Federated Learning as a Fairness Fix

The Expectation

‘FL improves diversity by training on data from everywhere.’



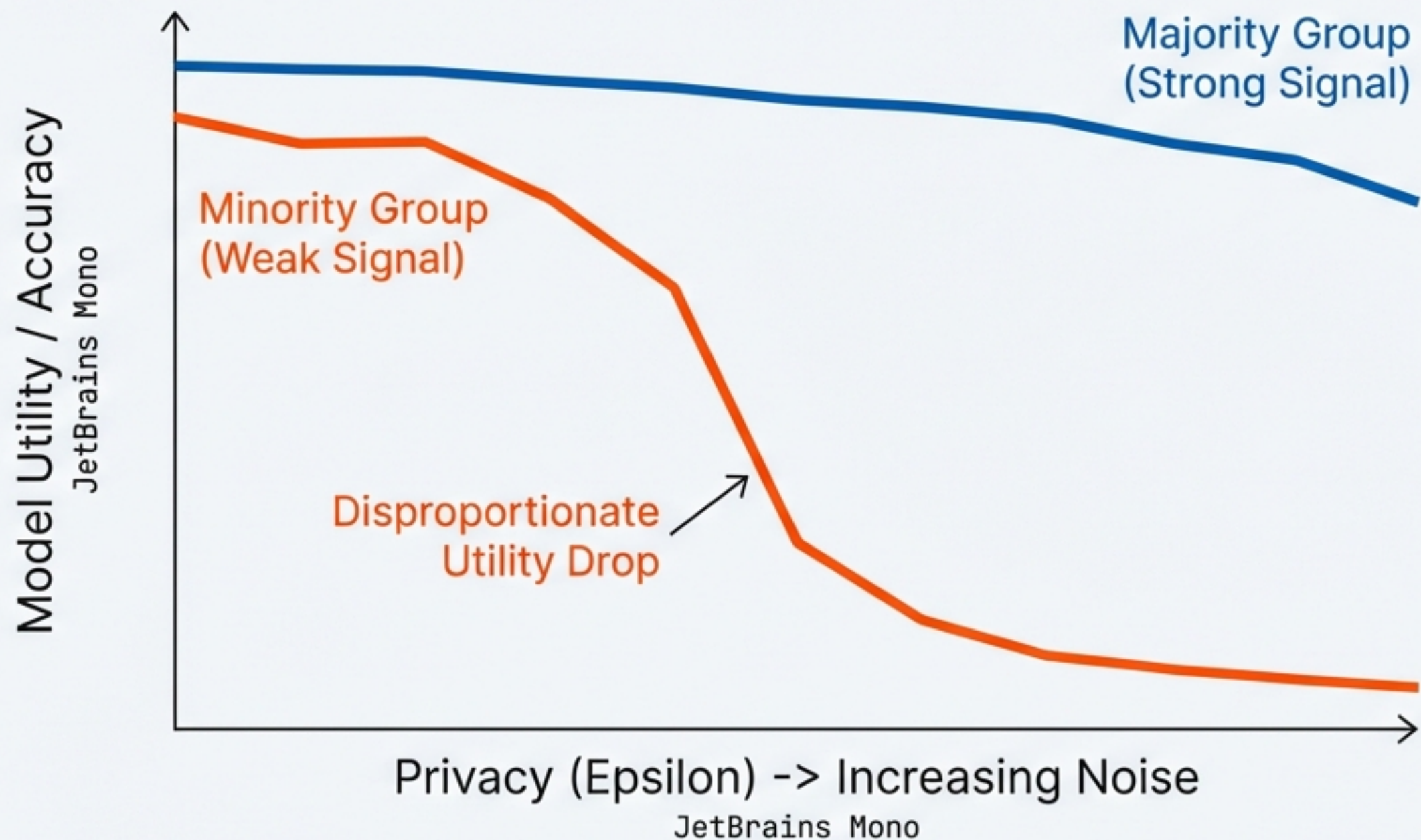
The Reality

1. **Participation Bias:** Users with unstable connectivity or old batteries drop out.
2. **Aggregation Bias:** Global models drift toward the majority device type.
3. **Non-IID Effects:** Differing demographics destabilize optimization.



Audit Action: Metrics must be evaluated per 'Client-Cluster' and 'Device Class'.

Privacy vs. Fairness: The Hidden Cost of Noise



The Mechanic: Differential Privacy adds noise.



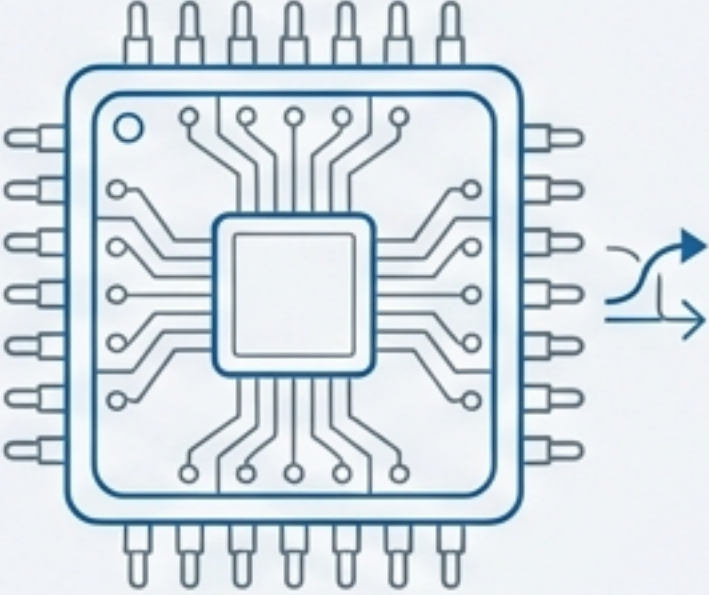
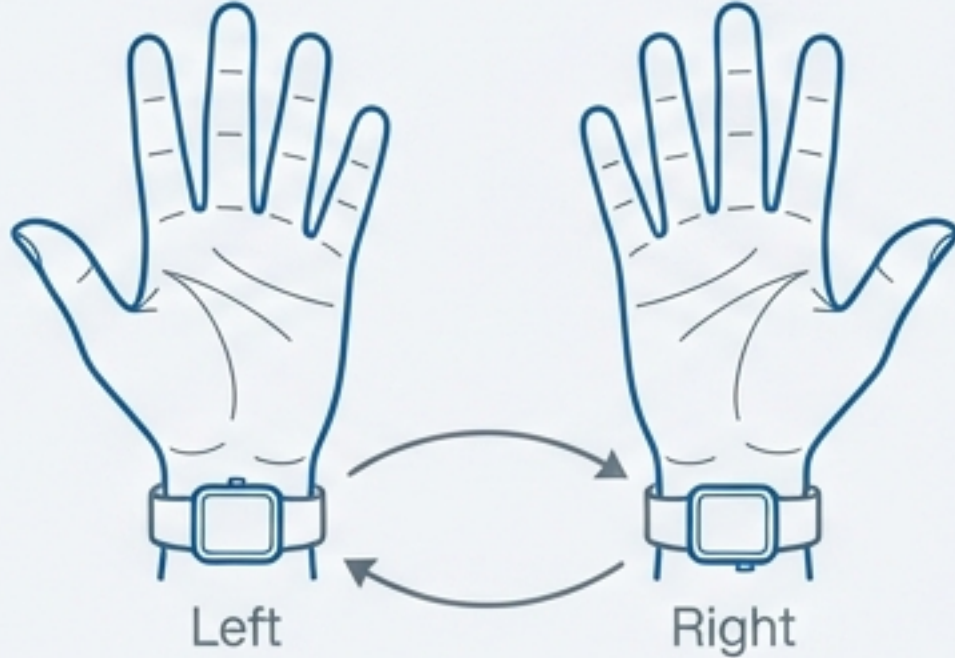
The Impact: Noise degrades “minority signals” (weaker physiological patterns) far more than majority signals.



Audit Action: Document privacy budgets explicitly. Do not assume DP is fairness-neutral.

Contextual Generalization is a Fairness Issue

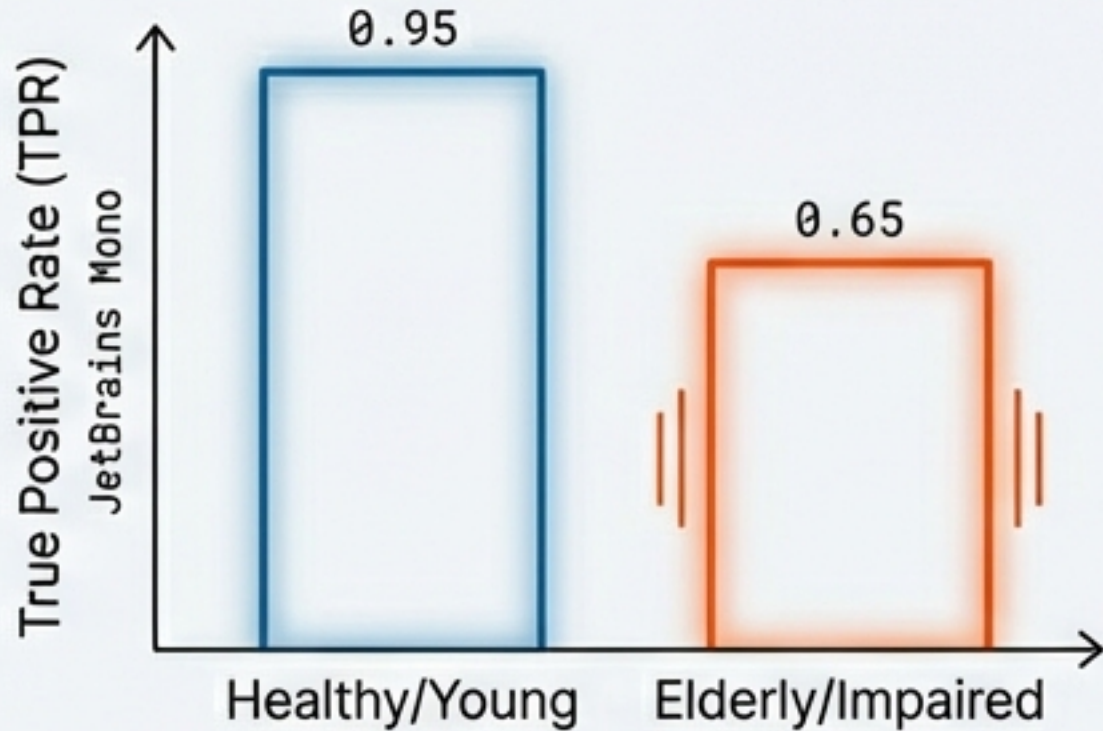
When the Model Leaves the Lab

1	Device Context	2	User Habit	3	Longitudinal Drift
					
Firmware changes or hardware generation gaps.		Switching from dominant to non-dominant wrist.		Aging or disease progression changing user baselines.	

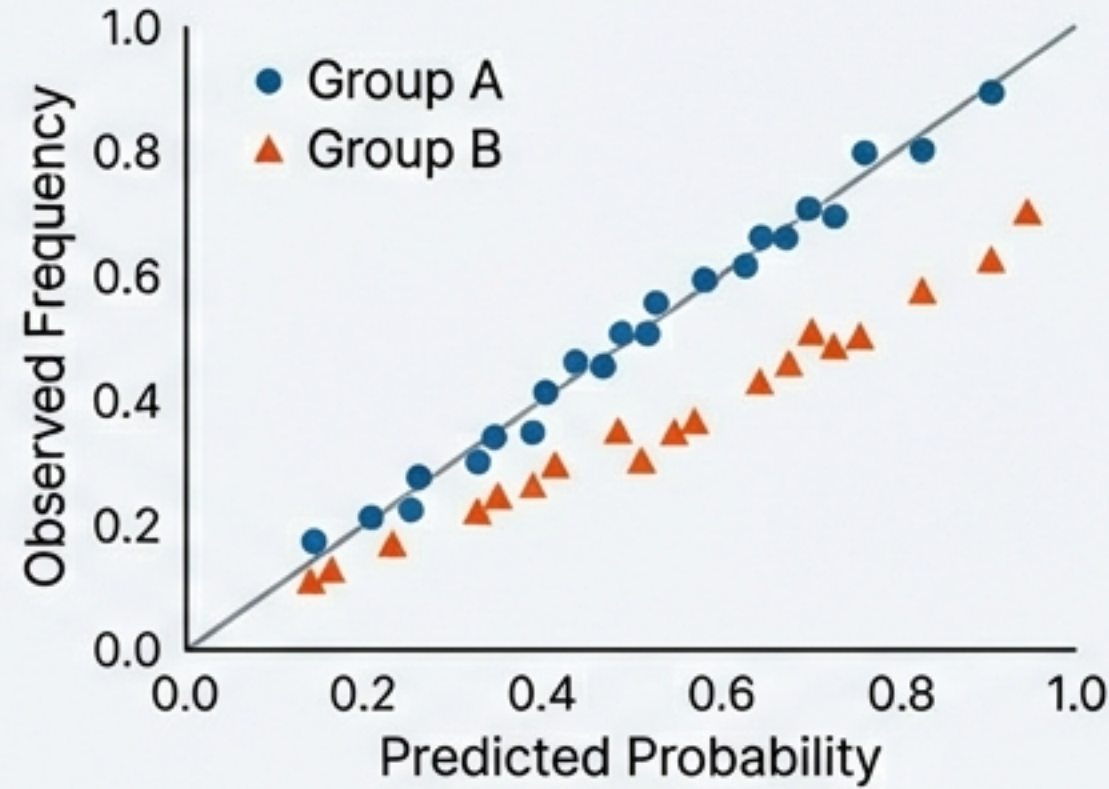
The Equity Angle: These failures cluster around users who cannot afford frequent device upgrades. Generalization failure is an economic bias.

Quantifying the Gaps: The Metrics Dashboard

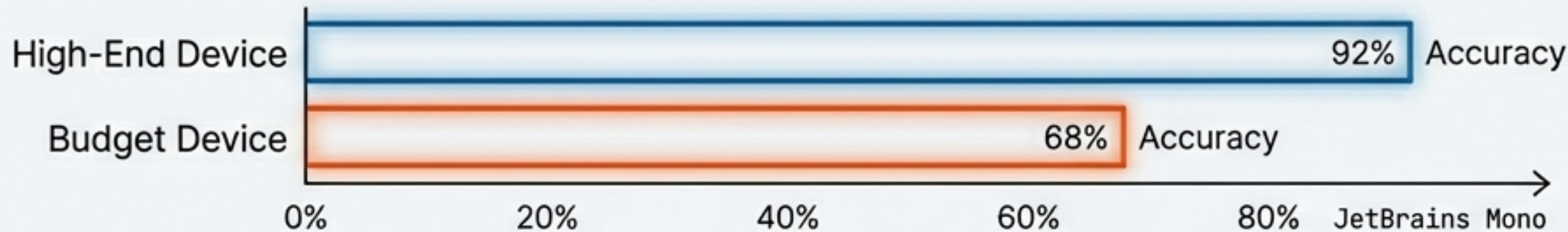
Intersectional TPR



Calibration Error



Device Class Degradation



Methodology:

Use **Bayesian** Credible Intervals for small sample groups.

Frequentist p-values ignore minority harms.

The Playbook in Action: Activity Recognition Audit

Case Study Findings

15%

True Positive Rate (TPR)
Disparity for elderly users
with mobility impairments.



Significant accuracy drop
on low-cost devices in
power-saving mode.

$\epsilon \leq 1$

Differential Privacy
amplified these gaps.

Mitigation Strategy: Task-aware reweighting in Federated Learning and device-adaptive inference thresholds.

Turning Compliance into a System Property

Legal Articles

Technical Implementations

GDPR Art. 5 (Fairness)  → Met via Documented Definitions

GDPR Art. 25 (Privacy by Design)  → Met via FL/DP Integration (Audited)

Data Minimization  → Met via Edge Inference

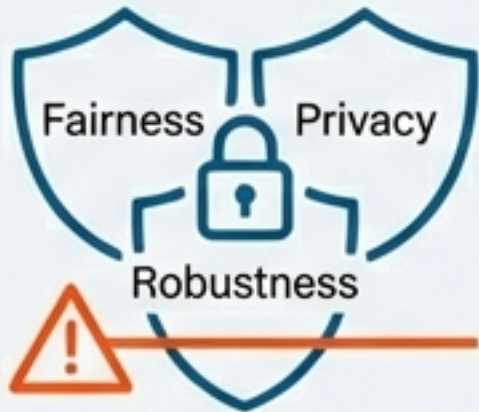
Rights of the Subject  → Met via Output Interpretability

Strategic Value: Fairness auditing moves from an ethical ‘add-on’ to a core accountability obligation.

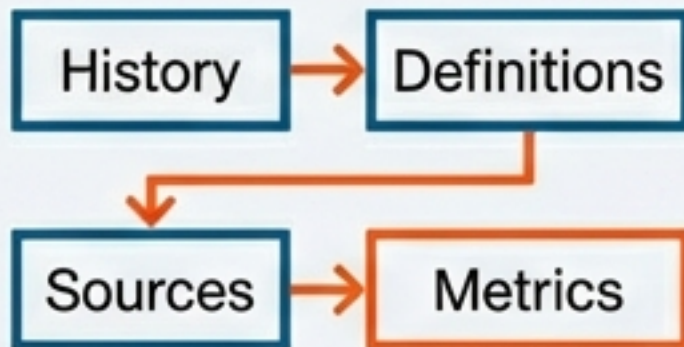
Moving Forward: Responsible Deployment



- Resource Reality: ~2-3 engineer days per module. Operationally feasible.



- Final Thesis: Fairness, Privacy, and Robustness are inseparable. FL and DP are insufficient without this audit layer.



- **Call to Action:** Integrate the 4-step workflow (History -> Definitions -> Sources -> Metrics) into the engineering lifecycle immediately.

Deploy Audit